

MathE Use Case

The MathE project (mathe.ipb.pt) is a digital platform designed to support the teaching and learning of mathematics in higher education. This platform provides personalized assessments and educational content across 22 higher education-level mathematical topics, supported by an extensive and complex database. For the purposes of the DataGEMS project, we will focus solely on the MathE functionalities necessary for the workflow of the project. Accordingly, we have developed an Entity-Relationship Diagram (ERD) that specifically addresses the tables relevant to the case study.

The following section presents the ERD of the database that stores this information, along with detailed descriptions of each attribute.

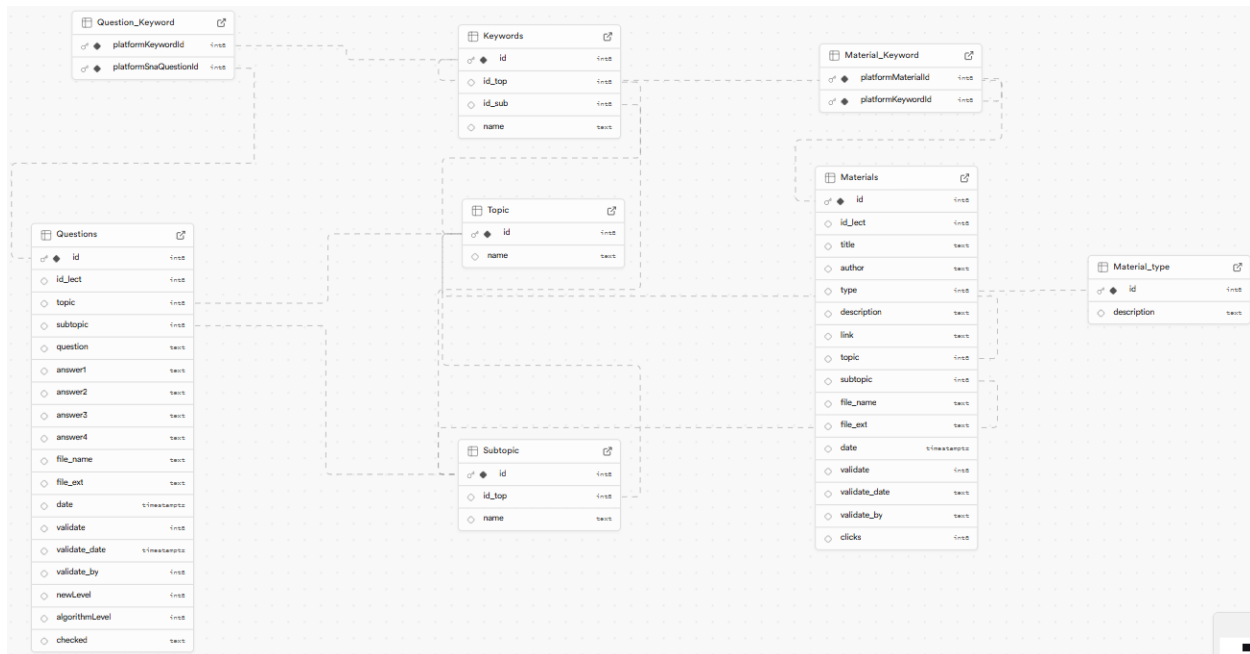


Figure 1 – ERD MathE Use Case (Supabase example)

<https://app.diagrams.net/?page-id=R2IEEEUBdFMjLlhIrx00&scale=auto>

1. Keywords:

- **Purpose:** Stores a list of keywords used to categorise questions and materials.
- **Attributes:**
 - **id** (Primary Key): Unique identifier for each keyword.
 - **id_top** (Foreign Key): A unique identifier from the Topic table.
 - **id_sub** (Foreign Key): A unique identifier from the Subtopic table.
 - **name:** The description of the keyword.

2. Materials:

- **Purpose:** Stores information about learning materials, such as documents or videos.
- **Attributes:**
 - **id** (Primary Key): Unique identifier for each material.
 - **id_lect** (Foreign Key): A unique identifier from the User table representing the Lecturer who created the material.
 - **title:** The title of the material.
 - **author:** The author of the material. (If it is a video or material created by someone else, it should be referenced here.)
 - **type** (Foreign Key): The material type (1 - Video Lessons, 2 - Video Review, 3 - Teaching Material).
 - **description:** A description of the material.
 - **link:** A link to the material (In the case of Video Lessons or Video Reviews).
 - **topic:** The topic of the material.
 - **subtopic:** The subtopic of the material.
 - **file_name:** The name of the associated file. (For written materials).
 - **file_ext:** The file extension.
 - **date:** The date the material was submitted.
 - **validate:** Validation flag. (1 - Accepted, 2 - Not Accepted, 3 - In Progress, 4 - Waiting for Validation).
 - **validate_date:** Validation date.
 - **validate_by** (Foreign Key): A unique identifier from the User table representing the Reviewer who validated the material.
 - **clicks:** A count of clicks or views for the material.

3. Material_Type:

- **Purpose:** Stores different types of learning materials (Video Lessons, Video Review, Teaching Material).
- **Attributes:**
 - **id** (Primary Key): Unique identifier for each material type.
 - **description:** The description of the material type.

4. Questions:

- **Purpose:** Stores information about questions. These are represented in the form of multiple-choice questions with four answer options or the option "I don't know."
- **Attributes:**
 - **id** (Primary Key): Unique identifier for each question.

- **id_lect** (Foreign Key): A unique identifier from the User table representing the Lecturer who created the question.
- **topic** (Foreign Key): A unique identifier from the Topic table.
- **subtopic** (Foreign Key): A unique identifier from the Subtopic table.
- **question**: The wording of the question in LaTeX.
- **answer1, answer2, answer3, answer4**: The answer options for the question. The **answer1** attribute always represents the correct answer.
- **file_name**: Some questions may include images or documents necessary for solving. This attribute represents the file name.
- **file_ext**: The file extension (JPEG, PNG, PDF, DOCX).
- **date**: The date the question was submitted.
- **validate**: Validation flag. (1 - Accepted, 2 - Not Accepted, 3 - In Progress, 4 - Waiting for Validation).
- **validate_date**: Validation date.
- **validate_by** (Foreign Key): A unique identifier from the User table representing the Reviewer who validated the question.
- **algorithmLevel**: The difficulty level associated with the question, assigned by the algorithm based on the history of assessments.

5. Material_Keyword:

- **Purpose**: This junction table handles the many-to-many relationship between the Materials and Keywords tables. A material can have multiple keywords, and a keyword can be associated with multiple materials.
- **Attributes**:
 - **platformMaterialId** (Primary Key & Foreign Key): A unique identifier that links to the Material table.
 - **platformKeywordId** (Primary Key & Foreign Key): A unique identifier that links to the Keyword table.

6. Question_Keyword Table:

- **Purpose**: This junction table handles the many-to-many relationship between the Questions and Keywords tables. A question can have multiple keywords, and a keyword can be associated with multiple questions.
- **Attributes**:
 - **platformQuestionId** (Primary Key & Foreign Key): A unique identifier that links to the Question table.
 - **platformKeywordId** (Primary Key & Foreign Key): A unique identifier that links to the Keyword table.